Round 3: what to run (Emory only)

One thing to run: a permutation test called "5d". Everything else is a short check. It takes about **4 hours on 4 cores** (Option A) or about **16 hours on 1 core** (Option B). Choose one option, not both.

Step 0. Copy the files (1 minute)

Copy these files from `diagnostics_code_Round3/1_Code/` **into the project's `1_Code/` folder**:

File	What it is
`03a_perm_null_r3.r`	the permutation run (new)
`fl_functions_r3.r`	the 5d shuffle (new)
`diag_functions_r3.r`	same as the Round 2 file, only faster (new)
`cleaning_functions_r2.r`	**unchanged** Round 2 file. If you still have it in `1_Code/`, you do not need to copy it again
`summarize_perm_r3.r`	makes the tables and the figure (new)
`run_5d_4cores.sh`	optional, for Option A on Mac or Linux

Packages: the same as before (`data.table`, `argparse`, `car`, `geepack`, `clubSandwich`). Nothing existing is overwritten. All output goes to a new folder:

`2_Pipeline/03-Run_regressions/Prot_multivar_diag_R3/`

How to run (every time): open the **Terminal** tab in RStudio, then go to the project folder (the same place where you run `03a`):

```
cd /path/to/31_NPS_proteomics_project_analysis
```

Use `emory` in lower case.

Step 1. Quick check (about 2 minutes)

```
Rscript 1_Code/03a_perm_null_r3.r --cohort emory --design fl --B 2 --nprot 30 --outtag R3smoke
```

Look at the console output:

- It prints a small table **"Subjects per Batch"**. **Please send me this table.**
- It must end with `Done: replicates 0 to 2 ....`
- If there is an error, or a warning that a Batch has fewer than 8 subjects, **stop and send me the last 20 lines**.

Step 2. The main run: choose Option A or Option B

Settings are the same in both options: B = 100 shuffles, 500 proteins.

Option A: four cores (about 4 hours)

We split the 101 replicates (0 to 100) into four blocks. Each block is a separate R process. **Check first that your computer has at least 4 free cores** (Mac: `sysctl -n hw.ncpu`; Linux: `nproc`). If not, use Option B or tell me.

A1. Mac or Linux: one command

```
bash 1_Code/run_5d_4cores.sh
```

It starts the four blocks by itself (the first one a minute ahead of the others) and waits. Messages of each block go to `logs_5d/block1.log` to `block4.log`.

A2. Any computer (Mac, Linux or Windows): four Terminal tabs

Open **four** Terminal tabs in RStudio (the + button next to the Terminal tab). In each tab go to the project folder (`cd ...`). Then paste **one** command per tab:

Tab 1 (start this one first, wait about one minute until you see `Proteins used: 500`):

```
Rscript 1_Code/03a_perm_null_r3.r --cohort emory --design fl --B 100 --nprot 500 --rep_from 0 --rep_to 1
```

Tab 2:

```
Rscript 1_Code/03a_perm_null_r3.r --cohort emory --design fl --B 100 --nprot 500 --rep_from 26 --rep_to
```

Tab 3:

```
Rscript 1_Code/03a_perm_null_r3.r --cohort emory --design fl --B 100 --nprot 500 --rep_from 52 --rep_to
```

Tab 4:

```
Rscript 1_Code/03a_perm_null_r3.r --cohort emory --design fl --B 100 --nprot 500 --rep_from 78 --rep_to
```

(Windows: the commands are the same. Type them on one line, without any \.)

When all four tabs print `Done:`, go to Step 3.

Option B: one core (about 16 hours)

One command, one tab:

```
Rscript 1_Code/03a_perm_null_r3.r --cohort emory --design fl --B 100 --nprot 500
```

Tips: start it in the evening. Keep the computer awake. On Mac or Linux you can run it in the background so that closing RStudio does not stop it:

```
nohup Rscript 1_Code/03a_perm_null_r3.r --cohort emory --design fl --B 100 --nprot 500 > run_5d.log 2>&1
```

Both options: if something stops

Run **the same command again**. Finished replicates are kept and skipped. To check the progress, count the files `*_repNNN.txt` in the output folder: 101 files means finished.

If you would like a shorter run first, use `--B 50` instead of `--B 100` in every command (about half the time). **Do not mix B values** in the same folder.

Step 3. Summary (about 1 minute)

Run this **after all 101 replicate files exist**:

```
Rscript 1_Code/summarize_perm_r3.r \
   --dir 2_Pipeline/03-Run_regressions/Prot_multivar_diag_R3/Perm5d_fl_protectyes \
   --full 2_Pipeline/03-Run_regressions/Prot_multivar_diag_R2/R2_IgnoreSVs/prot_multivar_ignoreSVs_emory
```

(On Windows put it on one line and remove the \.) The `--full` file is the Round 2 main run; it gives the red dots of all proteins. If you no longer have it, leave the `--full` part out.

Check: in `Perm5d_fl_protectyes_r3_table.txt`, the row `HTZ` should have `lambda_real` about **1.87** (the same as replicate 0 of Round 2). If it is not close to 1.87, **stop and tell me**.

Step 4. What to send back

All files are small and contain no individual-level data.

They are in `2_Pipeline/03-Run_regressions/Prot_multivar_diag_R3/Perm5d_fl_protectyes/`:

File	What it is
`*_manifest.txt`	the settings
`*_r3_table.txt`	**main table**: one row per reference method
`*_r3_by_d2.txt`	the same, by tertile of the denominator df
`*_r3_by_replicate.txt`	one row per replicate and method
`*_r3_qq_curves.txt`	numbers of the Q-Q curves
`*_r3_pvalues.txt`	p-values of all proteins and replicates (about 5 MB)
`*_r3_qq_2x3.png`	the Q-Q figure (2 x 3 panels)

Also send: - the **"Subjects per Batch"** table from Step 1, - the **time** the run took (the last lines of the log show it), - if there was an error: the last 20 lines of the console output.

Please **do not** send the large `*_repNNN.txt` files.

For me (not needed for running)

- Replicate 0 reproduces the real data: the script stops with an error if the rebuilt expression differs from the real one by more than 1e-8.
- The settings are the analysis settings: `--protect yes`, `--na zero` (missing set to 0 in the cleaning, as in script V), z-scaling, unstructured working correlation. The effect of missing values is a later question and is **not** tested in Round 3.
- Seed 20261011 and the same 500 proteins as Round 2.
- Test on synthetic data (never use for results): replicate 0 of 5d equals replicate 0 of 5b to 1e-15, and the p-values of `diag_functions_r3.r` equal those of `diag_functions_r2.r`.